

30) Show that $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$ are logically equivalent.

Soln: Let $\neg p \rightarrow (q \rightarrow r)$ be False. By defn. of conditional, $\neg p$ be True and $(q \rightarrow r)$ be False. By defn. of negation, $\neg(\neg p)$ is False. By double-negation law, $\neg(\neg p) \equiv p$. $\therefore p$ is False. By defn. of conditional, since $(q \rightarrow r)$ is False, $\therefore q$ is True and r is False.

Let $q \rightarrow (p \vee r)$ be False. By defn. of conditional, q is True and $(p \vee r)$ is False. By defn. of OR, both p, r are False.

$\therefore \neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$ are logically equivalent $\square$

31) Show that $p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$ are logically equivalent.

Soln: Let $p \leftrightarrow q$ be True. By defn. of biconditional, either both p, q are True or both p, q are False.

Let $(p \rightarrow q) \wedge (q \rightarrow p)$ be True. By defn. of AND, $p \rightarrow q$ is True and $(q \rightarrow p)$ is True. Let q be True. By defn. of conditional, since $(p \rightarrow q)$ is True, it is consistent. Since $(q \rightarrow p)$ is True, by defn. of conditional, p must be True.

$\therefore$ If q is True, then p is True.

Let q be False, $\therefore q \rightarrow p$ is True which is consistent. By defn. of conditional, since $p \rightarrow q$ is True, p must be False.

$\therefore$ If q is False, then p is False.

$\therefore p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$ are logically equivalent $\square$

32) Show that $p \leftrightarrow q$ and $\neg p \leftrightarrow \neg q$ are logically equivalent.

Soln: Let $p \leftrightarrow q$ be True. By defn. of biconditional, both p, q are True or both p, q are False.

Let $\neg p \leftrightarrow \neg q$ be True. By defn. of biconditional, both $\neg p, \neg q$ are True or both $\neg p, \neg q$ are False.

If $\neg p, \neg q$ are True, by defn. of negation, $\neg(\neg p)$ and $\neg(\neg q)$ are both False. By double negation law, $\neg(\neg p) \equiv p$ and $\neg(\neg q) \equiv q$. $\therefore$ Both p, q are False.